6.3 Object detection and bounding boxes

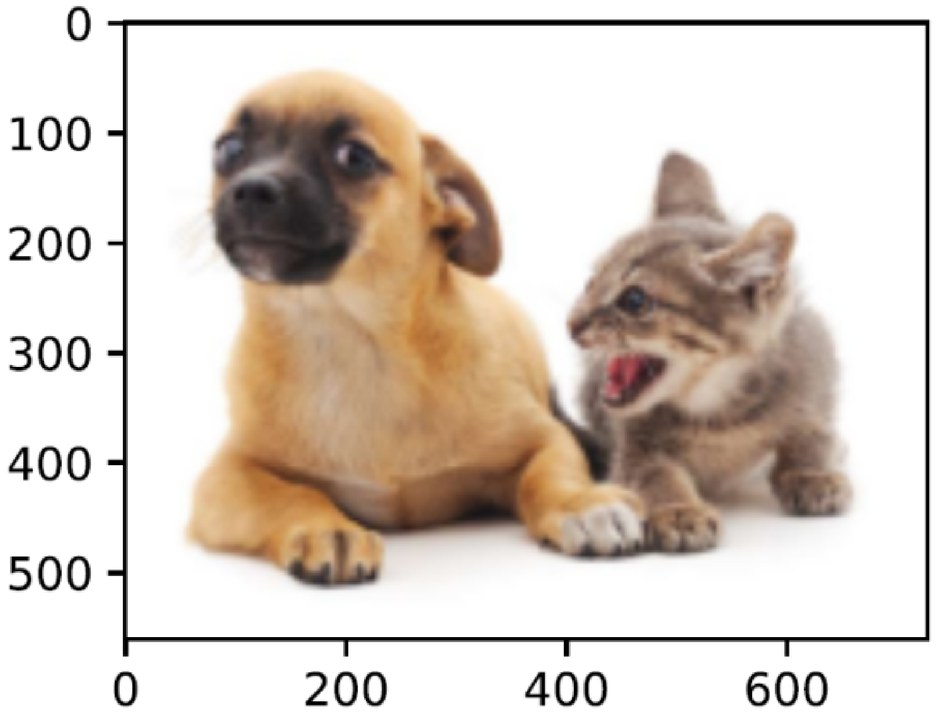
- in earlier sections (e.g., Section 3.7–Section 3.10), we introduced various models for image classification
- in image classification tasks, we assume that there is only *one* major object in the image, and we only focus on how to recognize its category
- however, there are often *multiple* objects in the image of interest
- we not only want to know their categories, but also their specific positions in the image
- in computer vision, we refer to such tasks as *object detection* (or *object recognition*)

6.3 Object detection and bounding boxes

- object detection has been widely applied in many fields
- for example, self-driving needs to plan traveling routes by detecting the positions of vehicles, pedestrians, roads, and obstacles in the captured video images
- besides, robots may use this technique to detect and localize objects of interest throughout their navigation of an environment
- moreover, security systems may need to detect abnormal objects, such as intruders or bombs

6.3 Object detection and bounding boxes

- in the next few sections, we will introduce several deep learning methods for object detection
- we will begin with an introduction to *positions* (or *locations*) of objects
- below we have the sample image to be used for object detection
- we can see that there is a dog on the left side of the image and a cat on the right; they are the two major objects in this image



6.3 Object detection and bounding boxes

- in object detection, we usually use a *bounding box* to describe the spatial location of an object
- the bounding box is rectangular, which is determined by the x and y coordinates of the upper-left corner of the rectangle and the x and y coordinates of the lower-right corner
- another commonly used bounding box representation is the (x, y)-axis coordinates of the bounding box center, and the width and height of the box
- we will define the bounding boxes of the dog and the cat in the image based on the coordinate information
- the origin of the coordinates in the image is the upper-left corner of the image, and to the right and down are the positive directions of the x and y axes, respectively

6.3 Object detection and bounding boxes

- let us draw the bounding boxes in the image
- after adding the bounding boxes on the image, we can see that the main outlines of the two objects are basically inside the two boxes

